SynthOmnicon Primitive Grammar

Operations & Algebra Reference Sheet

The 12 Primitives

Sym	Name	Values
D	Dimensionality	$D_\wedge, D_\triangle, D_\infty, D_{\text{holo}}$
T	Topology	$T_{\text{net}}, T_{\text{in}}, T_{\bowtie}, T_{\text{box}}, T_{\text{holo}}$
R	Recognition	$R_{\text{super}}, R_{\text{cat}}, R_\dagger, R_{lr}$
P	Polarity	$P_{\text{asym}}, P_\psi, P_\pm, P_{\text{sym}}, P_\pm^{\text{sym}}$
Γ	Int. Grammar	$\Gamma_{\text{and}}, \Gamma_{\text{or}}, \Gamma_{\text{seq}}, \Gamma_{\text{broad}}$
S	Stoichiometry	$1{:}1, n{:}n, n{:}m$
F	Fidelity	$F_\ell < F_{\text{eth}} < F_\hbar$
K	Kinetic Char.	$K_{\text{fast}} < K_{\text{mod}} < K_{\text{slow}} < K_{\text{trap}}$
G	Granularity	$G_\sqsupset < G_\rceil < G_\aleph$
H	Chirality	$H_0 < H_1 < H_2 < H_\infty$
Ω	Top. Protection	$\Omega_0 < \Omega_{Z_2} < \Omega_Z$
Φ	Criticality	$\Phi_{\text{sub}} < \Phi_c < \Phi_c^{\text{cx}} < \Phi_{\text{EP}} < \Phi_{\text{super}}$

Classes: categorical (6) — ordered (5) — special (Φ_c absorbing)

Distance Metric (§26.1)

Diagonal weighted Euclidean:

$$d(A,B) = \sqrt{\sum_{i=1}^{12} w_i (\xi_i^A - \xi_i^B)^2}$$

$\mathbf{w} = (1.0^{\times 9}, w_H{=}0.8, 1.0, w_\Omega{=}0.7)$

Riemannian (§26.2): $d_g = \sqrt{(\boldsymbol{\xi}^A - \boldsymbol{\xi}^B)^\top \Sigma^{-1} (\boldsymbol{\xi}^A - \boldsymbol{\xi}^B)}$

Directed: $d^{\rightarrow}(A \to B)_j = w_j \cdot \max(0, \xi_j^A - \xi_j^B)$ (upward free)

Core Operations

Meet $\sqcap$ (GLB)

Categorical: $A_i = B_i \to A_i$, else **CONFLICT**
Ordered: $\min(A_i, B_i)$
$\Phi_c \sqcap x = \Phi_c$

Join $\sqcup$ (LUB)

Categorical: $A_i = B_i \to A_i$, else **CONFLICT**
Ordered: $\max(A_i, B_i)$

Tensor $\otimes$ = Join with precondition:

$$A \otimes B = \begin{cases} A \sqcup B & \text{if no cat. conflicts} \\ \text{undefined} & \text{otherwise} \end{cases}$$

Physical: co-assembly requires categorical compatibility.

Algebraic Properties

- **Commutative**: $A \star B = B \star A$ for $\star \in \{\sqcap, \sqcup, \otimes\}$
- **Associative**: $(A \star B) \star C = A \star (B \star C)$ (when def.)
- **Idempotent**: $A \star A = A$
- Φ_c **meet-absorption**: $\Phi_c \sqcap x = \Phi_c$ for all x
- **Distance**: $d(A \otimes B, A)^2 \le d(A,B)^2$
- **Path-dep.**: $(A \otimes B) \otimes C$ may exist when $B \otimes C$ undefined

Cross-Primitive Axioms (Machine-Checked)

A1. $H_\infty \Rightarrow K \le K_{\text{trap}}$ (irrev. $\to$ trap)
A2. $\Omega \ge \Omega_Z \Rightarrow H \ge H_2$ (strong top. $\to$ chirality)
A3. $D_{\text{holo}} \Leftrightarrow T_{\text{holo}}$ (holo dim. $\leftrightarrow$ top.)
A4. $\Omega_Z \Rightarrow D_{\text{holo}}$ ($\mathbb{Z}$-protection $\to$ holo)

Path-Dependence Theorem

$(A \otimes B) \otimes C$ defined even when $B \otimes C$ undefined, iff $\exists$ ordered primitive i:

- $C_i > B_i$ (C needs more than B)
- $A_i = C_i$ (A supplies what C needs)
- No categorical conflict between $(A \otimes B)$ and C

Interpretation: Catalysis, scope promotion, chirality templating unified as "context A enables $B \to C$".

Mechanism	Blocking	A provides
Catalysis	Φ_c required	$A_\Phi = \Phi_c$
Scope promotion	$G_\aleph$ required	$A_G = G_\aleph$
Chirality templating	H_2 required	$A_H = H_\infty$

Primitive Semantics (Quick Ref.)

Categorical: exact match or **CONFLICT**
Ordered: min/max resolution
Φ_c: absorbing element

Structural Universality

Synthon: $\langle D; T; R; P; F; K; G; \Gamma; \Phi; H; S; \Omega \rangle$

Validity: Must satisfy axioms A1–A4.

Thesis: $d(\mathbf{x}, \mathbf{y}) = 0 \Rightarrow$ constraint-identical, substrate-independent.

Conflict Distance: $d_c(S) = \sqrt{|\text{Conf}(S)|}$ measures encoding reliability.

Dual Encoding:

- Holistic $\mathcal{E}_H(S)$: top-down from behavior
- Compositional $\mathcal{E}_C(S)$: bottom-up from $\bigotimes S_i$
- $\text{Conf}(S) = \{p : \mathcal{E}_H(S)[p] \ne \mathcal{E}_C(S)[p]\}$

Tensor Product Properties

When $A \otimes B$ defined:

- $(A \otimes B)_F = \max(F_A, F_B)$
- $(A \otimes B)_K = \max(K_A, K_B)$
- $(A \otimes B)_j = \max(A_j, B_j)$ for all ordered j
- Categorical: must agree (precondition)
- Φ_c absorption: if $A_\Phi = \Phi_c$ or $B_\Phi = \Phi_c$, then $(A \otimes B)_\Phi = \Phi_c$

Translation Cost: $\delta_A = d(A, A \otimes B)$, $\delta_B = d(B, A \otimes B)$; smaller δ is dominant.

Ouroboricity

Scalar: $\mathcal{O}(\mathbf{x}) = [\Phi{=}\Phi_c] \cdot (1 + [\Omega{\ne}\Omega_0] + [H{\ge}H_1] + [G{=}G_\aleph])$

Tiers (priority R1$\succ$R5):

- **R1** $\Phi_c + P_\pm^{\text{sym}} \to O_\infty$ (special Frobenius)
- **R2** $\Phi \in \{\Phi_{\text{sub}}, \Phi_{\text{super}}, \Phi_{\text{EP}}\} \to O_0$
- **R3** $\Phi_c + \Omega_0 \to O_1$
- **R4** $\Phi_c + \Omega{\ne}\Omega_0 + D \in \{D_\wedge, D_{\text{holo}}, D_\triangle\} \to O_2$
- **R5** $\Phi_c + \Omega{\ne}\Omega_0 + D_\infty \to O_2^\dagger$

Note: $\mathcal{O}$ cannot detect O_∞ (no P term); O_∞ absorbed by Φ_{EP} under $\otimes$.